

Economics - deals with study of production, distribution, exchange, and consumption of goods and services. 2 mark

Managerial economics - is the app. of economic theories, principles, and concepts to managerial decision making for solving ~~but~~ business problems and achieving organization objectives. 2 mark

"Economics is an Art" (10 mark) V-imp
 becoz it deals with practical application of economic ~~problem~~ principle to solve real life problems such as production, consumption, pricing and distribution of resources.
 It helps individuals, firm, & govt make efficient & rational decision.

Nature of managerial economics

1. **Apps of economic theory** - It applies economic concepts & principles to solve managerial decision making problem.
2. **Microeconomic in nature** - as it studies individual firms, pricing, cost & p/p decision.
3. **Decision oriented** - focuses on making rational & optimum decisions for business management.
4. **Practical approach** - emphasizes practical apps rather than pure theory.

Scope of managerial economics.

1. Demand analysis and forecasting - analysis of consumer demand & production prediction of future demand.
2. Production & cost analysis.
3. Pricing decision - ~~price~~ fixing prices under diff market condⁿ.
4. Profit management - planning, forecasting, and controlling profits.

Basic economic activities of an economy.

6 marks

↳ are activities carried out to satisfy human wants ~~and~~ using scarce resources

1. Production - creating or adding utility to goods and services to satisfy human wants
ex - manufacturing goods
2. Consumption - use of goods & services to satisfy human wants
ex - using food, clothes electricity
3. Exchange - buying & selling of good & services through money
- it connects producer & consumer in the market

Consumer goods & capital good are both final goods

+ consumer good satisfy direct wants

+ while capital goods help in further production & are treated as final goods in national income accounting.

Deductive Method- starts from general assumptions & proceeds to draw specific conclusions through logical reasoning.

• economic laws are derived on the basis of assumptions like rational b^{eh} behavior of consumers & other factors remaining constant.

ex- law of demand.

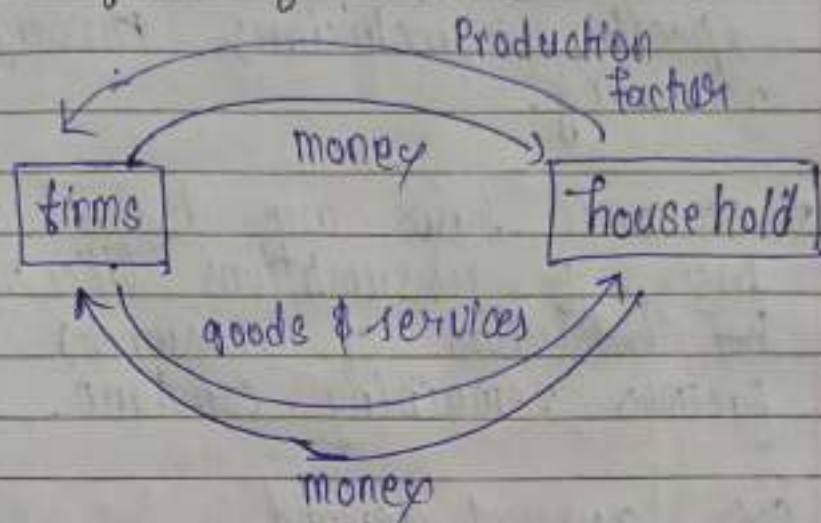
Inductive method- starts from particular facts & ob^s observations and leads to general conclusion.

• economic laws are formulated after collecting & analyzing real-life data & statistical studies.

ex- economic generalization

4. Distribution - refers to sharing of national income among diff. factors of production such as wages, rent, interest, and profit.

* Circular flow of economic activities.



* Economic problem - arise b'coz human wants are unlimited while resources are scarce and have alternative uses.

3 main economic problem.

- What to produce & how much to produce?
- How to produce?
- For whom to produce?

* Consumer good
used for consumption
ex- food, clothes

Capital good
used for further production
ex- machine tool.

*) Consumer goods are final goods which are directly consumed by consumers.

R mark

National Income - is the total monetary value of all final goods & services produced by the ~~res~~ residents of a country during a given period of time (1yr)

Measurement of national income.

1. Product Method - is measured by adding the value added by all productive units.
2. Expenditure method - measured by adding final expenditure on goods & services.

3 Income method - estimates national income by adding all factor income earned by ~~non-res~~ residents during a year.

Steps:

1. Identification of factor income (wages, rent interest profit)
2. Add mixed income of self employed person
3. Calculation of net domestic income at factor cost.
4. Addition of net factor income ~~from~~ from abroad.

(2) Precaution while estimating national income by income method.

→ Exclude transfer income. (ex-pensions, scholarships, gifts & unemployment allowance)

→ Avoid double counting of income.

- only factor income.
- income arising from the sale of second-hand good ~~must~~ be excluded.

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* Various concepts of National Income

1. ~~Nation~~ Gross domestic product - (GDP)

- is the total monetary value of all final goods and services produced within the domestic territory of a country during a year.

2. Net Domestic Product (NDP)

- after deducting depreciation from GDP.

i.e. $NDP = GDP - \text{depreciation}$.

3. Gross National Product - (GNP)

.... including net factor income from abroad, during a year.

i.e. $GNP = GDP + \text{Net factor income from abroad}$.

4. Net National ^{product} income (NNP)

$NNP = GNP - \text{depreciation}$.

5. Personal Income - is the total income actually received by individual & households from all sources during a year, before payment of personal taxes.

6. Disposable Income - is the income left with an individual after payment of personal taxes which

$DI = PI - \text{taxes}$.

7. Private Income - is the total income earned by private sector, including individuals & private enterprises, from both factor income and transfer payments during a given period.
8. Per capita Income - avg. income of country's population.

$$PCI = \frac{\text{national income}}{\text{total population.}}$$

Q What does the circular flow model depicts?

- 2 mark
- continuous flow of good & services & money income betⁿ household and firms in an economy.
 - It shows how production, income, consumption and expenditure are interrelated

Demand

V. Imp
2 mark Law of Demand - states that other things remaining constant, when price of a commodity rises, its quantity demanded falls. and when the price decreases, its quantity demand rises.

Thus price and quantity demanded have an inverse relationship.

Notes

$$\text{price} \propto \frac{1}{\text{quantity demanded}}$$

2 mark Demand - refers to the quantity of a commodity which consumers are willing and able to buy at a given price, during a given period of time.

Factors which lead to "↓ in demand"

- 4 mark
1. ↑ in price of commodity - higher price reduces purchasing power, so demand falls.
 2. ↓ in income of consumer - with lower income consumer buys less
 3. ↑ in price of substitute goods - consumer shift to substitute.

2 mark Elasticity of demand - measures the responsiveness of demand to change in price, income, or price of related goods.

2 mark
* Price elasticity of demand - refers to the degree of responsiveness of quantity demanded to change in price of a commodity, other factors remaining constant.

i.e. it shows how much demand changes due to change in price.

4 mark
* Degree of price elasticity of demand

1. Perfectly elastic demand ($E_d = \infty$)

- small change in price causes infinite change in demand

2. Perfectly inelastic demand ($E_d = 0$)

- demand does not change with price.

3. Relatively elastic demand ($E_d > 1$)

- % change in demand is greater than price change.

4. Relatively inelastic demand ($E_d < 1$)

- % change in demand is less than price change.

Note: $E_d = 1$ $\rightarrow$ unitary elastic demand.

1. Survey method - demand is conducted by conducting surveys of buyers.

2. Expert opinion method - demand is estimated by expert opinion.

3. Statistical method - uses past data & trends to predict future demand.

4. End-user method - demand is estimated by studying present & future uses of the product.

* Demand forecasting - means the estimation or prediction of future demand for a product over a given period of time.

* Types of demand -

1. Price demand - quantity of a commodity demanded at a different price, other things remaining constant.

2. Income demand - shows relationship between income of consumer & quantity demanded of a product.

3. Cross demand - demand for a commodity in relation to the change in price of another related commodity.

4. Joint demand - when two or more goods are demanded together to satisfy single want.

* determinant of demand

1. Price of commodity
2. Income of consumer
3. Price of related goods
4. Taste & preferences of consumers
5. No. of consumers.

* Demand f^n - shows functional relationship betn quantity demanded & its determinant.

$$Q_d = f(P, Y, T, P_r, N)$$

quantity demanded

P - price of commodity
 Y - income
 T - taste good
 P_r - Price of related ↑
 N - No of consumer.

* Determinants of demand forecasting.

1. Past demand trends
2. Income changes
3. Govt policy & control
4. Business & economic condⁿ

Supply - supply refers to the quantity of a commodity which producers are willing & able to offer for sale at a given price, during a given period of time, other things remaining constant.

* determinants of supply • price of commodity

- cost of production
- Govt policy.
- Technology.

2 mark *

- Elasticity of supply. - refers to degree of responsiveness of quantity supplied to a change in price of a commodity, other factors remaining constant.

$$E_s = \frac{\% \text{ change in quantity supply}}{\% \text{ change in price}}$$

- Q1 When the price of a commodity is ₹20 per unit, its quantity demanded is 800 units. When price rises by ₹5 per unit, quantity given: Original Price $P_1 = ₹20$ demanded falls by 20%. Calculate price of elasticity of demand. Also state whether demand is elastic or inelastic. give reasons.

Given: Original price $P_1 = ₹20$

rise in price = ₹5

New price $P_2 = ₹25$

Original quantity demanded $Q_1 = 800$ units

Fall in quantity = 20% of 800

$$= \frac{20}{100} \times 800 = 160 \text{ unit.}$$

New quantity demand (Q_2) = $800 - 160 = 640$ units

price elasticity of demand = ?

$$\text{Price elasticity of demand (Ed)} = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

here it is $Q_1 - Q_0$ bcoz quantity changed is given in % in question otherwise always take $Q_1 - Q_0$

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$$\% \text{ change in quantity} = \frac{Q_1 - Q_0 \times 100}{Q_1} = \frac{800 - 640 \times 100}{800} = 20\%$$

$$\% \text{ change in price} = \frac{P_2 - P_1 \times 100}{P_1} = \frac{25 - 20 \times 100}{20} = 25\%$$

Now,

$$E_d = \frac{20\%}{25\%} = 0.8$$

- ∴ Price elasticity of demand (E_d) = 0.8, which is less than 1
→ demand - inelastic.

Reason: % fall in quantity demanded (20%) is less than % rise in price (25%)

Q2. Price of good A rises from ₹ 20 to ₹ 30 quantity supplied rises from 200 units to 800 units. Calculate elasticity of supply.

given: Original price $P_1 = ₹ 20$

New price $P_2 = ₹ 30$

$$\% \text{ change in price} = \frac{P_2 - P_1 \times 100}{P_1} = \frac{30 - 20 \times 100}{20} = 50\%$$

original quantity $Q_1 = 200$

New quantity $Q_2 = 800$

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$$\% \text{ change in quantity} = \frac{Q_2 - Q_1}{Q_1} \times 100$$

$$= \frac{800 - 200}{200} \times 100$$
$$= 300\%$$

Now, elasticity of supply

$$E_s = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

$$= \frac{300\%}{50\%} = 6 \rightarrow \text{elastic supply}$$

Note

$E_s/E_d = \infty \rightarrow$ Perfectly elastic

$E_s/E_d = 0 \rightarrow$ Perfectly inelastic

$E_s/E_d = 1 \rightarrow$ Unitary elastic

$E_s/E_d > 1 \rightarrow$ elastic

$E_s/E_d < 1 \rightarrow$ inelastic

Q Calculate and comment.

10 mark

1. The price of tea per cup is $\downarrow$ sed from Rs 4 to Rs 3 and the demand of coffee is $\uparrow$ sed from 2 cups per day to 4 cups per day. Calculate cross elasticity of demand.

given:- Price of tea (P_t) = ₹4

New price (tea) $P_0 = ₹3$

demand of coffee (Q_1) = 2 cups/day

New demand (coffee) $Q_2 = 4$ cups/day

$$\frac{Q_2 - Q_1}{Q_1} \times 100 = \frac{4 - 2}{2} \times 100$$

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$$E_c = \frac{\% \text{ change in quantity of coffee}}{\% \text{ change in price of tea}}$$

$$E_c = \frac{100}{-25} = -4$$

$$\frac{P_2 - P_1}{P_1} \times 100 = \frac{3 - 4}{4} \times 100 = -25\%$$

1) Tea & coffee are substitute goods

2. Mr. Gupta income is raised from Rs 10,000 to Rs 15,000 and demand for good A is raised from 500 to 800 units. Calculate income elasticity of demand.

$$\% \Delta Q = \frac{800 - 500}{500} \times 100 = 60\%$$

$$\% \Delta Y = \frac{15000 - 10000}{10000} \times 100 = 50\%$$

$$E_y = \frac{\% \Delta Q}{\% \Delta Y} = \frac{60}{50} = 1.2$$

1) $E_y > 1$, elastic demand-

3. The demand of commodity X is raised from 200 to 250 units when price rises from ₹5 to ₹6. Calculate price elasticity of demand.

$$\% \Delta Q = \frac{250-200}{200} \times 100 = 25\%$$

$$\% \Delta P = \frac{6-8}{8} \times 100 = -25\%$$

$$E_d = \frac{25}{-25} = -1$$

ignore -ve sign here
↓ only in price elasticity.

$$\therefore E_d = 1$$

demand is unitary elastic

4. IF the price rises of a good A rises from ₹20 to ₹30. Its supply increases from 200 to 800 unit. Calculate elasticity of supply.

$$\% \Delta Q = \frac{800-200}{200} \times 100 = 300\%$$

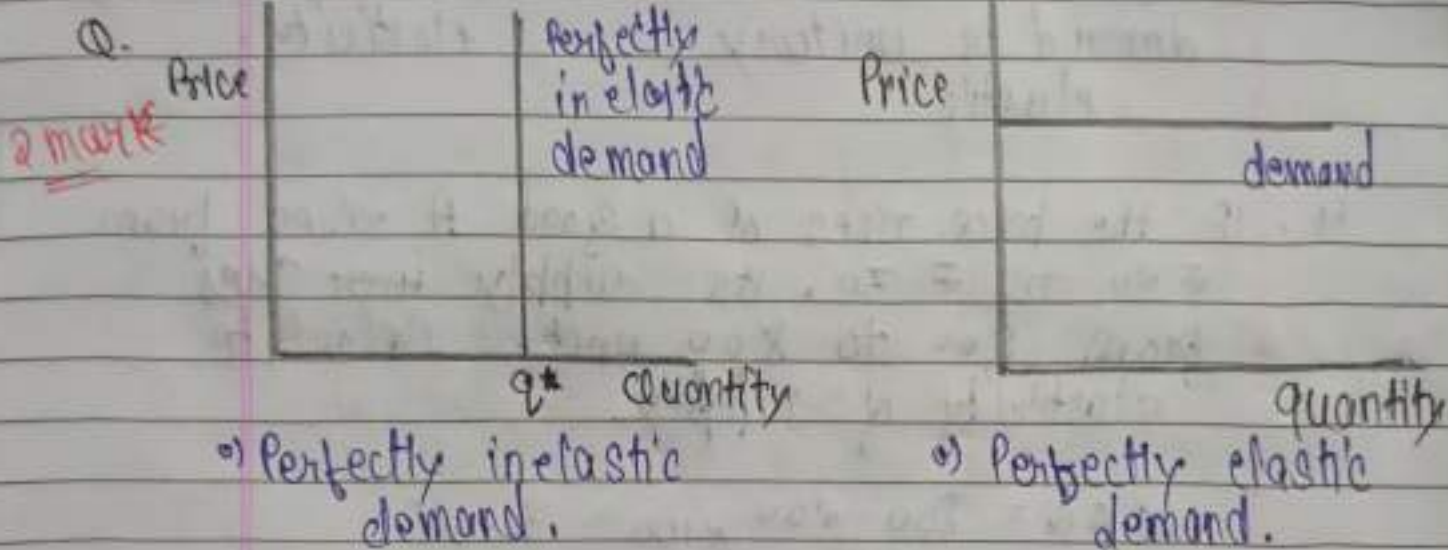
$$\% \Delta P = \frac{30-20}{20} \times 100 = 50\%$$

$$E_s = \frac{300}{50} = 6 > 1 \quad [\text{supply highly elastic}]$$

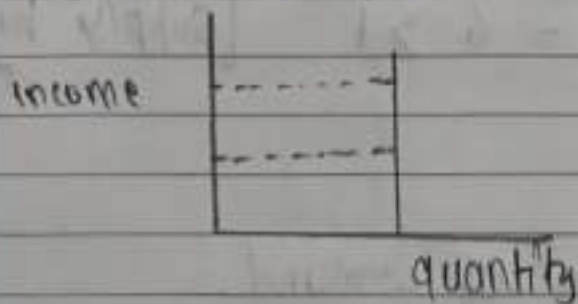
Note: Cross elasticity of demand.

(E_c) → +ve → substitute good
 -ve → complementary good
 0 → unrelated good.

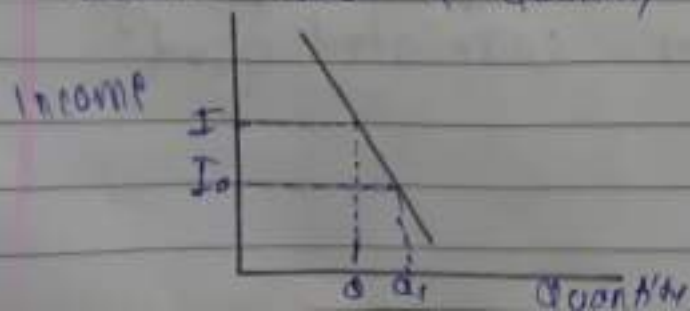
Q. Griffen Paradox - refers to the exceptional situation in which demand for a commodity rises when its price rises and falls when its price falls, thus violating the law of demand.



1. Zero income elasticity - ($E_y = 0$) - when income demand changes but quantity demanded remains unchanged.

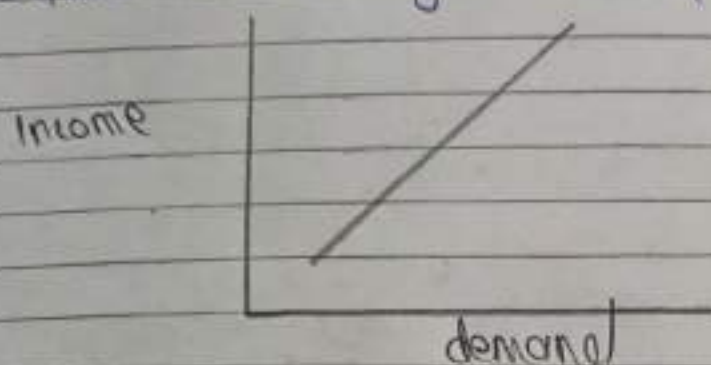


e. -ve income ~~it~~ elasticity ($E_y < 0$) - when income rises & quantity demanded falls.

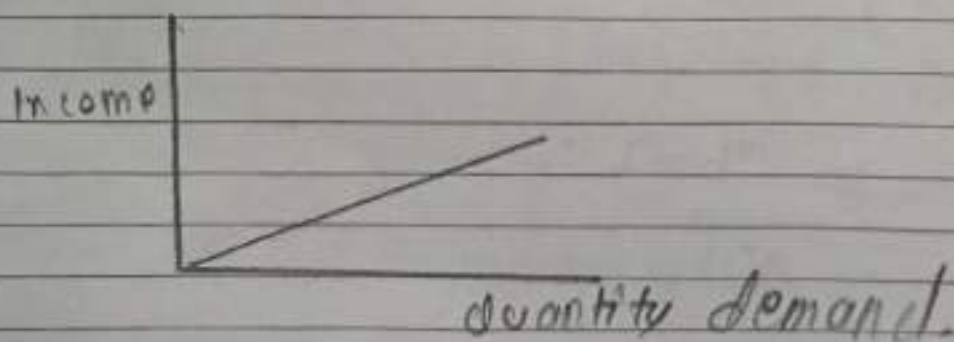


3.

Unit income elasticity of demand. ($E_y = 1$)
when % change in income causes an
equal % change in quantity demanded



4. Income elasticity greater than unity ($E_y > 1$)
when % change in quantity demand is
greater than % change in income.



UNIT - 3

1 imp
2 mark Production fⁿ - shows the functional relationship betⁿ i/p and o/p produced during a given period of time.
 ⇒ It shows max o/p that can be produced with the given set of i/p and technology.

2 mark Marginal product of an i/p - refers to the additional o/p produced by employing one extra unit of a variable i/p (labour), while other i/p are kept constant

$$\text{Marginal Product (MP)} = \frac{\Delta \text{ total product (TP)}}{\Delta \text{ units of i/p}}$$

Cobb-Douglas production fⁿ (Mathematical form)

2 mark

$$Q = AL^{\alpha}K^{\beta}$$

Q = o/p, L = Labour i/p

K = capital i/p, A = technology const.

α, β = o/p elasticities of labour & capital

1 mark Least cost combⁿ of i/p - refers to the combⁿ of different factors of production that produces a given level of o/p at the minimum possible cost.
 i.e. using i/p in most economic way.

⇒ Condⁿ of least cost combⁿ

$$\frac{MP_L}{P_L} = \frac{MP_K}{P_K}$$

MP_L, MP_K = marginal product of labour & capital

P_L, P_K = Price of labour & capital

UNIT-3 (Production)

V.V./mp Law of Variable Proportions - states that
10 marks "when one factor of production is used while other factors are kept const. & total o/p first rises at increasing rate, then rises at diminishing rate, and finally rises."

1. Stages of Law of variable proportions Increasing return

- TP rises at an increasing rate.
- MP rises
- fixed factors are better utilised

This stage is not suitable for production b'coz resources are under utilised.

2. Diminishing Return

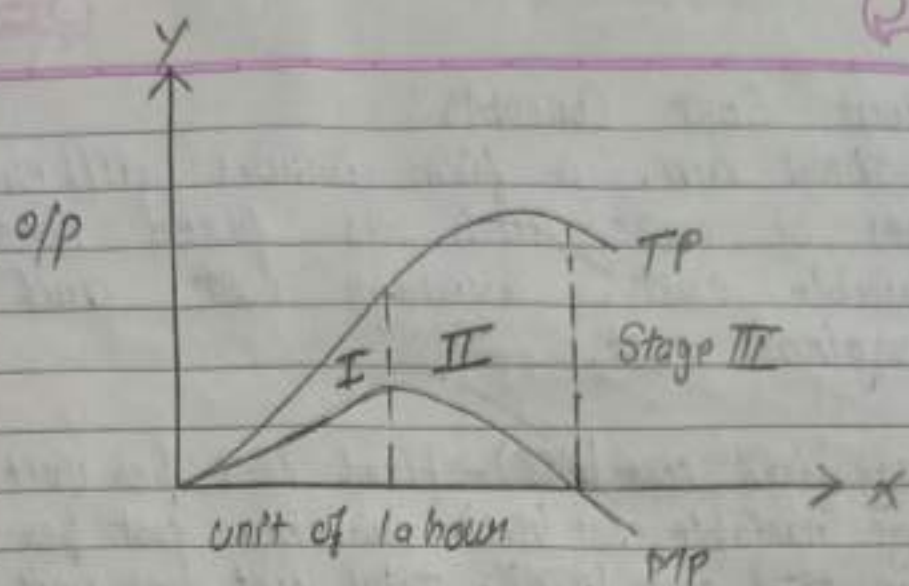
- TP rises ~~at~~ but at diminishing rate.
- MP starts falling but remains +ve.
- fixed factors become over-crowded.

This is the most efficient & rational stage of production

3. Negative Returns

- TP declines
- MP become -ve
- Too much labour cost inefficiency.

Production should never operate in this stage.



Stage I : TP rising fast , MP rising

Stage II : TP rising slowly , MP falling

Stage III : TP falling , MP $\rightarrow$ -ve

Explicit Cost - are the actual cash payments made by a firm to outsiders for using factors of production.

ex- wages paid to workers, interest paid on loan , cost of raw material.

Implicit cost - are the opportunity costs of using owners own resources in the business , for which no cash payment is made.

ex- interest on owners own capital.

Various Cost Concepts:

Imp
10 marks
4
In short run, a firm incurs different types of cost such as fixed cost, variable cost, average cost and marginal cost.

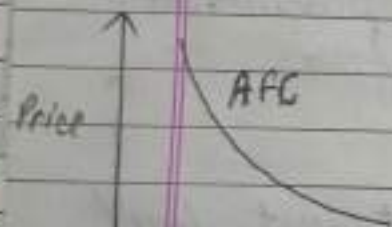
Average fixed cost (AFC) - fixed cost per unit of o/p

Average variable cost (AVC) - variable cost per unit of o/p.

Average total cost (ATC) - total cost per unit of o/p.

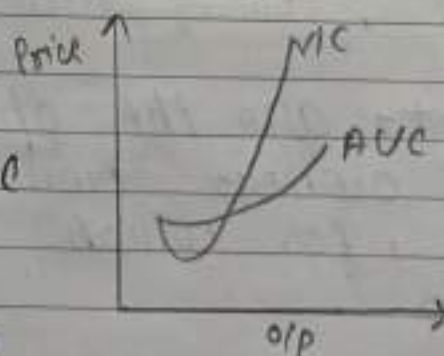
$$ATC = AVC + AFC$$

(1) Relationship betⁿ AFC & o/p



→ AFC continuously falls as o/p rises
b'coz fixed cost is spread

(2) Relation betⁿ AVC & MC



$MC < AVC \rightarrow$ AVC falls

$MC > AVC \rightarrow$ AVC rises

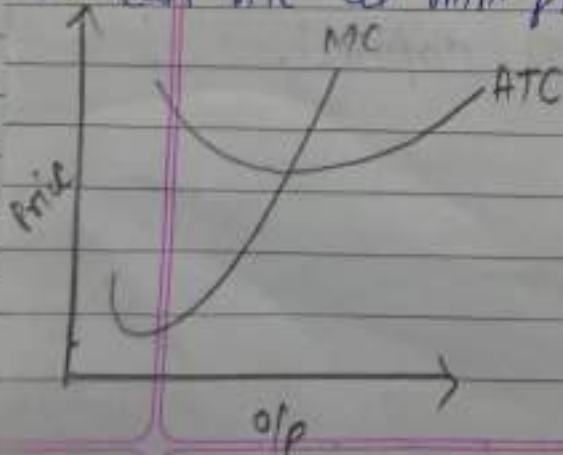
→ MC cuts AVC at min. pt.

(3) Relation betⁿ ATC & MC

$MC < ATC \rightarrow$ ATC falls

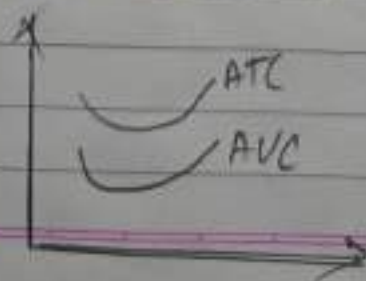
$MC > ATC \rightarrow$ ATC rises

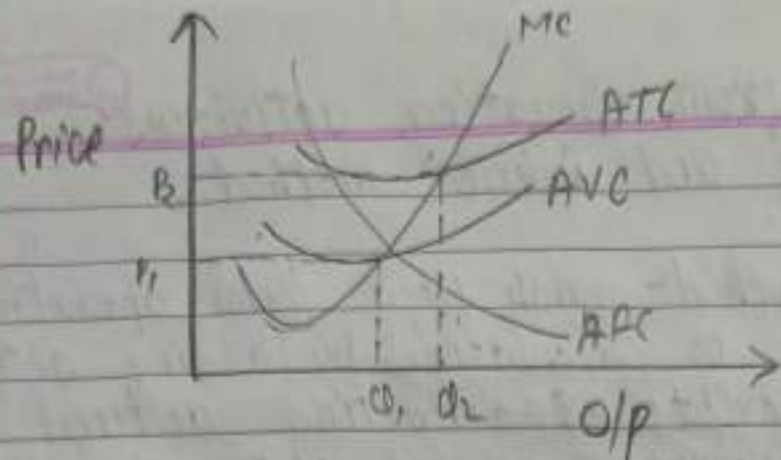
MC cuts ATC at min. pt.



(4) Relation betⁿ ATC & AVC

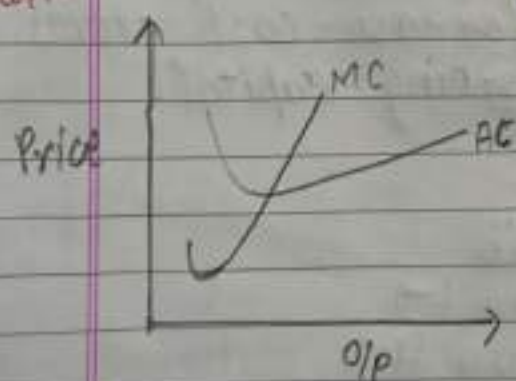
ATC is always greater than AVC b'coz it includes AFC





- Thus, the various cost concepts are closely related.
- MC governs the behaviour of both AVC & ATC, while AFC continuously declines.
- These relationships help the firm to determine the optimum level of Q/p .

* Avg. cost (AC) & MC $\rightarrow$ additional cost of producing 1 more unit of Q/p .
 i.e. cost per unit of output



$MC < AC$, AC falls
 $MC > AC$, AC rises

MC cuts AC at min. pt.

* Opportunity Cost - is the value of the next best alternative foregone when a choice is made among scarce resources.

ex- if a student chooses to study instead of working, wages foregone are the opportunity cost.

Cashflow from operating activities by direct and indirect method.

1. Direct Method - cash flow from operating activities is calculated by adding actual cash receipts & deducting actual cash payments - related to operations

ex: Cash received from customer = 5,00,000
cash paid to suppliers & employees = 3,50,000
income tax = 30,000

$$\begin{aligned}\text{Net cash from operating activities} &= 5,00,000 - 3,50,000 - 30,000 \\ &= 1,20,000\end{aligned}$$

2. Indirect Method - ... is calculated by adjusting net profit for non-cash items and changes in working capital.

ex: Net profit = 1,00,000
add depreciation = 20,000
inc in debtors = 10,000 (-)
inc in creditor = 5,000 (+)

$$\begin{aligned}\text{Net cash from operating activities} &= 1,00,000 + 20,000 - 10,000 + 5,000 \\ &= 1,15,000\end{aligned}$$

Economies of scale - refers to the reduction in avg cost of production due to the increase in scale of op.

ex- A large firm buys raw materials in bulk & uses specialised machines, which reduces cost per unit.

Diseconomies of scale - arise when an increase in scale of production leads to an increase in avg. cost.

ex- In a very large firm, management & coordination problem increases the cost per unit.

Stock	Flow
measured at a pt of time	measured over a period of time.
ex- capital on 31st march	ex- income earned during a year

Production possibility Curve - shows the different combinations of goods that an economy can produce with given resources & technology in a given period of time.

Dividend payout ratio = shows the portion of net profit distributed to shareholders as dividends.

$$\text{dividend payout ratio} = \frac{\text{dividend per share}}{\text{earning per share}} \times 100$$

Q

Given below is the cost schedule of a firm.
Its avg. fixed cost is Rs 20 when it produce

3 units

Q (units)	1	2	3
AVC (Rs)	20	28	32

Soln.

$$ATC = AVC + AFC$$

$$TC = ATC \times Q$$

$$MC = ATC$$

calculate its MC & ATC
at each given level
of Q

Step 1

Q	AVC	AFC	ATC	TC
1	20	20	50	50
2	28	20	48	96
3	32	20	52	156

Step 2

Q/P Change	ATC	MC
0-1	0 → 50	50
1-2	50 → 96	46
2-3	96 → 156	60

Step 3

$$Q=1 \rightarrow ₹ 50 \rightarrow ATC$$

$$Q=2 \rightarrow ₹ 48$$

$$Q=3 \rightarrow ₹ 52$$

$$Q=1 \rightarrow 50 \rightarrow MC$$

$$Q=2 \rightarrow 46$$

$$Q=3 \rightarrow 60$$

Following table gives the total cost schedule of the firm. It is also given

that the AFC is 4 units of o/p is ₹ 5

Q (Quantity)	TC
1	50
2	65
3	75
4	95
5	130
6	185

Find TVC & TFC of the firm for the corresponding values of o/p

Given: AFC at 4 unit = ₹ 5

$$TFC = AFC \times Q = 4 \times 5 = 20$$

TFC remains const. at all o/p levels.

Q	TC	TFC	TVC = TC - TFC
1	50	20	30
2	65	20	45
3	75	20	55
4	95	20	75
5	130	20	110
6	185	20	165

UNIT-4 (Market Structure)

Oligopoly - is a market structure in which few firms dominate the market and sell homogeneous or differentiated products.

*) Firms are interdependent, so the decision of one firm affects others.

*) Characteristic

+ Few sellers dominate market.

+ Firms are interdependent.

+ Homogeneous or differentiated product

*) ex- petroleum, steel, cement industry.

10 # Perfect competition - is a market structure in which there are large no. of buyers & sellers, selling homogeneous products, with free entry and exit of the firm.

*) All buyers and sellers have perfect knowledge of market price, so a single uniform price prevails in the market.

*) Firms are price takers, not price makers.

*) Characteristic

+ Large no. of buyers & sellers.

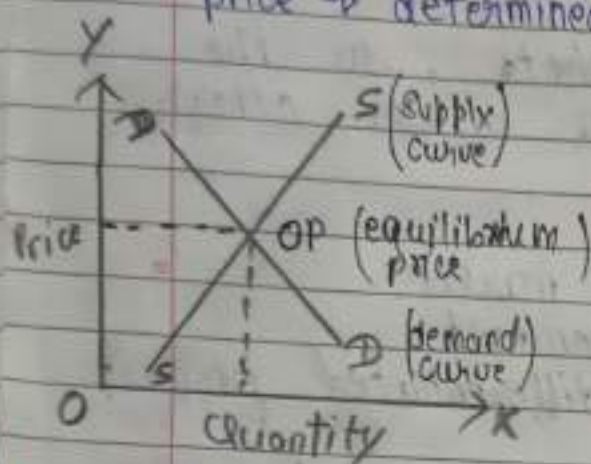
+ Homogeneous product.

+ Free entry & exit of firm

+ Perfect knowledge of price.

* Price & o/p determination under perfect competition

1. Price determination
price $\rightarrow$ determined at industry level

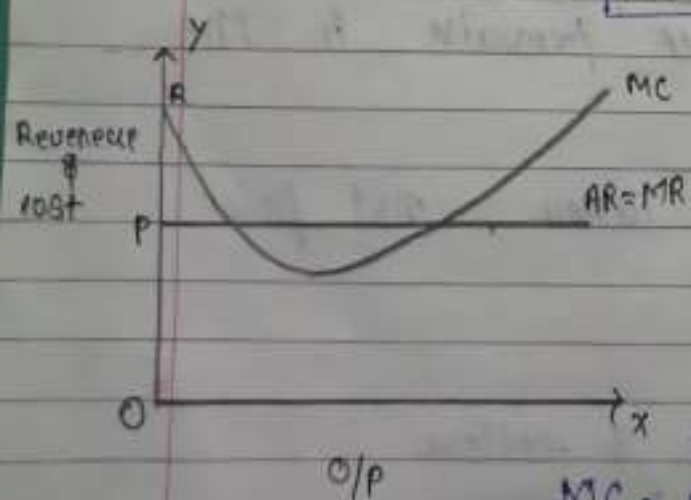


- * Industry demand curve (DD) slopes downward.
- * Industry supply curve (SS) slopes upward.
- * pt. where $DD = SS$, determines an equilibrium price (Op)

2. Output determination
individual firm $\rightarrow$ price taker
 $\hookrightarrow$ accepts market price determined by industry

$\therefore$ for a firm, $AR = MR = \text{Price}$.

- $\Rightarrow$ Firm decides o/p based on profit maximization condⁿ i.e. $MR = MC$



$\hookrightarrow$ at this pt.

- MC cuts MR from below
- Profit is max. or loss is min.

MR $\rightarrow$ marginal revenue
MC - marginal cost
AR - avg. revenue
AC - avg cost
AVC - avg variable cost

Situations of price and op determination.

- ① Short Run - firm may earn
- + Supernatural profit - when $\text{price} > AC$
 - Normal profit - $\text{price} = AC$
 - Loss - $\text{price} < AC$

- in short run - no. of firms is fixed & firms cannot change plant size.
- price $\rightarrow$ industry
- op $\rightarrow$ individual firm } determined.

② Long run - free entry & exit of firm is possible.

- supernatural profit $\rightarrow$ attract new firms.
- losses $\rightarrow$ force inefficient firms to leave.
- finally, all firms earn normal profit.
- condⁿ in long run
 $\text{price} = AR = MR = MC = AC$

Q. "Under perfect competition, the seller is price taker". explain (2 marks)

- b'coz there are large no. of seller & product is homogeneous.
- Market price is determined by industry demand & supply, no seller can influence it.
 - Therefore, seller accepts given price & adjust only op

- Q. "A competitive firm is not a price maker, but adjuster". Explain with reference to price-determination in long & short term. **10 MARK**

Under perfect comp., firm is not a price maker but a price adjuster becoz market price is determined by industry demand & supply.

- Individual firm has no control over price and therefore adjust o/p level acc. to given price in both run (short & long)

- Q1 Competitive firm cannot influence price due to:
- large no. of buyers & sellers.
 - Homogeneous product.
 - perfect knowledge of price.

- Q2 explain price determination for short & long run both.

**2 MARK** Monopoly - is a market structure in which there is a single seller producing a commodity for which there are no close substitutes.

The monopolist has full control over supply & therefore can influence the price.

2 MARK

- Q1 That is why in monopoly, seller is price maker.
- Q2 Features / characteristics **2 MARK**
- single seller & large no. of buyers
 - No close substitute.
 - High barrier to entry
 - price maker

ex - Indian railways

Monopolistic Market - is a market structure in which many sellers sell differentiated product which are close substitute for each other.

each firm has limited control over price due to product differentiation

Characteristic - large no. of seller.

- 1) differentiated product
- 2) free entry & exit of firm.

2/4 marks # diff. market structure.

Oligopoly	Perfect comp.	Monopoly	Monopolistic.
1) few sellers	large no. of sellers sellers.	single sellers	Many seller.
2) Homogenous or differentiated product	Homogeneous product	no close substitute.	differentiated product
3) Restricted	Free entry & exit	Highly restricted	Free entry exit
4) considerable influence.	Price taker	Price maker	limited price control
ex - steel, cement, petroleum	ex - agriculture product.	ex - Indian railway	ex - soap, toothpaste

Q. Why demand curve of a firm under monopolistic competition more elastic than under monopoly? 2 mark
 becoz in monopolistic comp.

- availability of close ~~sub~~ substitute = many firms sell differentiated products
- If one firm raises price, consumer can easily shift to other brand.
- limited price control.

while in monopoly

- no close substitute
- consumer have no alternative
- demand is less elastic.

a. Freedom of entry exit in monopolistic competition. explain. 2 mark

1. New firms can freely enter the industry if existing firms earn supernatural profit
2. loss making firms can freely leave industry w/o restriction.

beco of this - in long run - firms earn only normal profit

→ Competition remains strong.

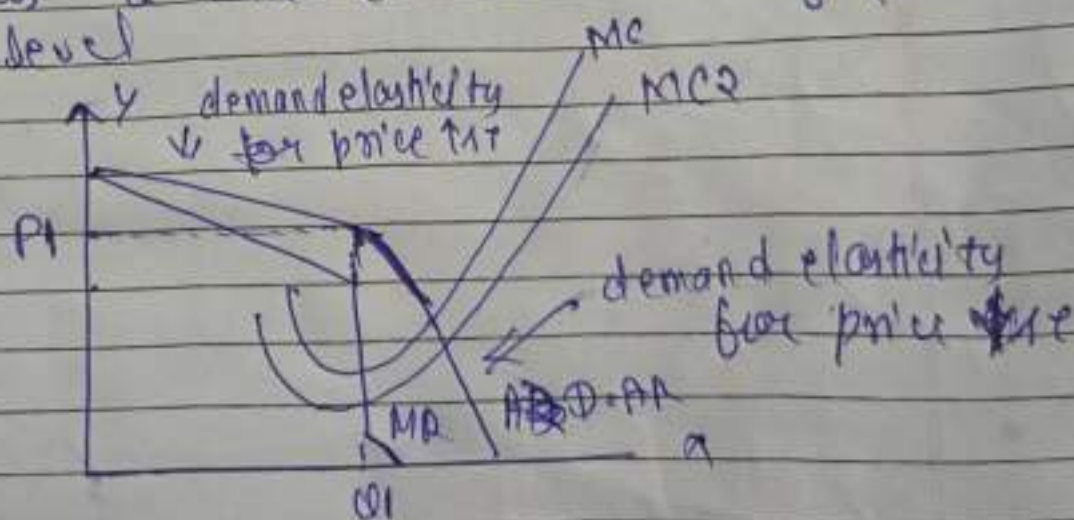
Q

Kinked demand curve under oligopoly? 14
 ↳ explains price rigidity.

Answer

- If a firm raises price
- other firms do not follow price rise
 - The firm loses customers
 - demand above kink (and) is highly elastic
- If a firm lowers price
- other firms follow the price
 - no significant gain in sales.
 - demand below the kink is less elastic

b'coz of this behaviour, demand curve has a kink at the existing price level



Capital Budgeting - refers to the process of evaluating and selecting long-term investment project whose benefits are expected to be received over many years.

ex - purchase of a machinery, plant expansion.

* Payback period - is the time required to recover the initial investment of a project from its cash inflows.

$$\text{Payback period} = \frac{\text{Initial Investment}}{\text{annual cash inflow}}$$

Capital Budgeting technique - used to evaluate profitability and feasibility of investment project.

A Traditional Method

1. Payback period method - measures how quickly investment is recovered.

merit - simple to calculate

demerit - ignores time value of money

ex investment = 1 lac, cash inflow = 20k
payback period = 5yr.

2. Accounting rate of return (ARR)

$$ARR = \frac{\text{avg profit}}{\text{avg. investment}} \times 100$$

merit - $\Rightarrow$ Focuses on accounting profit
demerit - $\Rightarrow$ ignores cash inflows

B. Modern Methods

1. Net present value (NPV)

$$NPV = \text{present value of cash inflow} - \text{initial investment}$$

merit $\left\{ \begin{array}{l} \Rightarrow \text{accept project if } NPV > 0 \\ \Rightarrow \text{consider time value of money.} \\ \Rightarrow \text{most reliable method,} \end{array} \right.$
demerit $\Rightarrow$ complex calculation

2. Internal rate of return (IRR)

$\Rightarrow$ is the rate at which $NPV = 0$

merit - time value considered, demerit - multiple IRR problem

3. Profitability Index (PI)

$$PI = \frac{\text{PV of cash inflow}}{\text{initial investment}}$$

$\Rightarrow$ accepts if $PI > 1$

$\Rightarrow$ Modern technique are superior as they consider time value for money.

Balance Sheet - is an important financial statement that shows the financial position of a business on a particular date.

→ It represents what business owns and what it owes.

→ It shows assets and liabilities of a business at a given pt. of time.

→ Acc to accounting principle,

$$\boxed{\text{Assets} = \text{Liabilities} + \text{Capital}}$$

Hence, both sheets of the balance sheets are always equal.

→ Nature / characteristic

- It is a statement, not an account.
- It shows financial position, not profit & loss.
- It is prepared on a specific date.

→ Structure of balance sheet - balance sheet has 2 sides.

1. Liability side - shows what business owes.
Includes - capital, long term liability (loans), current liability.

2. Assets side - shows what business owns
includes - fixed asset (land, building),
- current assets (cash, stock)

DATE: _____
PAGE NO: _____

1) Format of balance sheet.

Balance sheet as on _____	
Liabilities	Assets
Capital	Land & buildings
Long-term loans	Machinery
Bill payable	stock

Imp -> shows financial position of a business.

2) helps in decision making for investors.

* Q. mark
V. Imp
Liabilities - are the financial obligations or debts of a business which required to pay to outsider in future.

ex- loans

* Assets - are resources owned by the business which have economic value & are expected to provide future benefits.

ex- cash, stock, machinery, building,

* Financial Statement Analysis - is the process of examining financial statement to assess the financial performance & financial position of a business.

2 mark
* Profit & Loss statement - is a financial statement that shows net profit or net loss of a business for a specific accounting period by matching revenues with expenses.

2 mark
* Fund flow statement - is a statement that shows the sources and uses of funds of a business during a particular period.

2 mark
* Financial accounting - is the branch of accounting that deals with recording, classifying, & summarizing financial transactions of a business & presenting them through financial statements for external users.

V-imp # Ratio Analysis - is a technique of financial statement analysis in which relationship betⁿ 2+ financial figures related is expressed in form of ratios to evaluate financial performance & position of firm

→ Objective / Imp.

1. Helps in assessing earning capacity of business.
2. Helps in judging long term financial stability.
3. Useful for managers, investors, & creditors in taking financial decision.

* 2 mark Liquidity ratio - measures the ability of a firm to meet its short-term obligations using its current assets.

$$\text{Liquidity ratio} = \frac{\text{current asset}}{\text{current liability}}$$

* 2 mark Solvency Ratio - measures the ability of a firm to meet its long-term liabilities & shows long term financial stability of the business.

$$\text{Solvency ratio} = \frac{\text{total debt}}{\text{share holder equity}}$$

* Debt turnover ratio - measures the efficiency of a firm in collecting cash from its debtors during a particular period.

$$\text{debt \& turnover ratio} = \frac{\text{Net credit sales}}{\text{avg. debtors}}$$

* Golden rule of accounting for real accounts

a mania $\rightarrow$ debit what comes in &
credit what goes out.